

★ Read the questions carefully  
 ★ Make sure that you answered everything

**Problem 1.** Use the regression results below to answer the questions.  $lprice$  log(price, \$1,000),  $lsqrft$  log(size of house in square feet),  $llotsize$  log(size of lot in square feet), *colonial* colonial style (=1), *bdrms* number of bedrooms

. reg lprice lsqrft llotsize colonial bdrms

| Source   | SS         | df | MS         | Number of obs | = | 88     |
|----------|------------|----|------------|---------------|---|--------|
| Model    | 5.20397919 | 4  | 1.3009948  | F(4, 83)      | = | 38.38  |
| Residual | 2.81362433 | 83 | .033899088 | Prob > F      | = | 0.0000 |
|          |            |    |            | R-squared     | = | 0.6491 |
|          |            |    |            | Adj R-squared | = | 0.6322 |
| Total    | 8.01760352 | 87 | .092156362 | Root MSE      | = | .18412 |

| <u>lprice</u>   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|-----------------|-------------|-----------|-------|-------|----------------------|-----------|
| lsqrft          | .7071931    | .092802   | 7.62  | 0.000 | .5226138             | .8917725  |
| llotsize        | .1678189    | .0381807  | 4.40  | 0.000 | .0918791             | .2437587  |
| <u>colonial</u> | .0537962    | .0447732  | 1.20  | 0.233 | -.035256             | .1428483  |
| bdrms           | .0268305    | .0287236  | 0.93  | 0.353 | -.0302995            | .0839605  |
| _cons           | -1.349589   | .651041   | -2.07 | 0.041 | -2.644483            | -.0546947 |

(a) Interpret the coefficient on  $lsqrft$

log-log      A 1% increase in the size of the house increases the housing price by 0.71%, all else equal.

(b) Interpret the coefficient on *colonial*

log-lin      Housing prices are 5.38% higher for colonial-style houses compared to non-colonial style houses, all else equal. However, the estimate is not statistically sig. at the 5% lv.

(c) Interpret the coefficient on *bdrms*

log-lin      An add'l bedroom increases housing prices by 2.68%, all else equal. However, the estimate is not statistically sig. at the 5% lv.

**Problem 2.** Based on the table below, answer the questions.

| Dependent Variable=Labor Force Participation |                                       |                    |                    | ↓                  | ↓                  |
|----------------------------------------------|---------------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                              |                                       | (1)                | (2)                | (3)                | (4)                |
|                                              |                                       | LPM                | Probit             | LPM                | Probit             |
| $\beta_1$                                    | HHI                                   | -0.11**<br>(0.006) | -0.38**<br>(0.019) | -0.09**<br>(0.006) | -0.32**<br>(0.021) |
| $\beta_2$                                    | LF experience                         |                    |                    | -0.01**<br>(0.000) | -0.03**<br>(0.001) |
| $\beta_3$                                    | nchild<6yrs old                       |                    |                    | -0.14**<br>(0.005) | -0.49**<br>(0.016) |
| $\beta_4$                                    | nchild 6~18yrs old                    |                    |                    | -0.02**<br>(0.003) | -0.09**<br>(0.010) |
| $\beta_5$                                    | Husband's earned income<br>(\$10,000) |                    |                    | -0.01**<br>(0.001) | -0.02**<br>(0.004) |
| $\beta_0$                                    | Constant                              | 0.83**<br>(0.004)  | 0.97**<br>(0.016)  | 1.09**<br>(0.008)  | 1.97**<br>(0.039)  |
|                                              | Observations                          | 22,369             | 22,369             | 22,259             | 22,259             |
|                                              | R-squared                             | 0.017              |                    | 0.071              |                    |

Robust standard errors in parentheses

\*\* p<0.01, \* p<0.05, + p<0.1

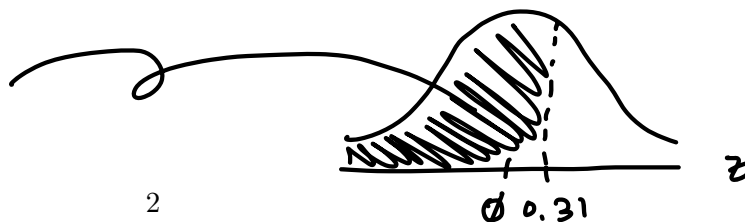
$$P(LFP=1) = \beta_0 + \beta_1 HHI + \beta_2 exp + \beta_3 nch6 + \beta_4 nch18 + \beta_5 hinc + u$$

- (a) Suppose a married woman is covered by her husband's insurance (HHI), has 10 years of working experience, and 2 children who are younger than 6 years old and her husband's income is \$30,000. In column (3), what is the probability that the woman will participate in the labor force?

$$\begin{aligned}
 P(LFP=1 \mid HHI=1, exp=10, nchild<6=2, nchild6-18=0, hinc=3) \\
 &= 1.09 + (-0.09) + (-0.01) \cdot 10 + (-0.14) \cdot 2 + 0 + (-0.01) \cdot 3 \\
 &= 0.59
 \end{aligned}$$

- (b) In column (4), what is the probability that the woman will participate in the labor force?

$$\begin{aligned}
 P(LFP=1 \mid HHI=1, exp=10, nchild<6=2, nchild6-18=0, hinc=3) \\
 &= \Phi(1.97 - 0.32 + (-0.03) \cdot 10 + (-0.49) \cdot 2 + 0 + (-0.02) \cdot 3) \\
 &= \Phi(0.31) \\
 &= 0.6217
 \end{aligned}$$



→ I added this question to help you review Q2 in Discussion 3.

**Problem 3.** Using data from a private school voucher program, also known as a school choice program, we estimated the following regression. *choicelyrs* represents the number of years a student attended a choice school from 1991 to 1994. Since it is a categorical variable that takes the values 0, 1, 2, 3, and 4, we generated binary variables *choicelyrs1*, *choicelyrs2*, *choicelyrs3*, and *choicelyrs4*.

```
. reg mathscore choicelyrs1 choicelyrs2 choicelyrs3 choicelyrs4 black hispanic female
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| Source   | SS         | df  | MS         | Number of obs | = | 990    |
|----------|------------|-----|------------|---------------|---|--------|
| Model    | 37872.1524 | 7   | 5410.30749 | F(7, 982)     | = | 13.52  |
| Residual | 392958.98  | 982 | 400.161894 | Prob > F      | = | 0.0000 |
|          |            |     |            | R-squared     | = | 0.0879 |
|          |            |     |            | Adj R-squared | = | 0.0814 |
| Total    | 430831.132 | 989 | 435.622985 | Root MSE      | = | 20.004 |

| mathscore   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|-------------|-------------|-----------|-------|-------|----------------------|-----------|
| choicelyrs1 | .5102603    | 1.735531  | 0.29  | 0.769 | -2.895516            | 3.916037  |
| choicelyrs2 | -1.314817   | 2.098575  | -0.63 | 0.531 | -5.433025            | 2.803392  |
| choicelyrs3 | -2.766788   | 2.198078  | -1.26 | 0.208 | -7.080257            | 1.546681  |
| choicelyrs4 | -.617641    | 2.856311  | -0.22 | 0.829 | -6.222817            | 4.987535  |
| black       | -16.19045   | 1.849133  | -8.76 | 0.000 | -19.81916            | -12.56174 |
| hispanic    | -13.37447   | 2.40842   | -5.55 | 0.000 | -18.10072            | -8.648231 |
| female      | 1.370047    | 1.277174  | 1.07  | 0.284 | -1.136258            | 3.876351  |
| _cons       | 57.04585    | 1.660486  | 34.35 | 0.000 | 53.78734             | 60.30436  |

- (a) Interpret the coefficient on *choicelyrs1* (omitted baseline = *choicelyrs* 0)

Students who attended a choice school for one year

scored 0.51 points higher in math than those who never attended, all else equal.

- (b) How would you rewrite a regression model to examine whether math scores differ between students who have ever attended a choice school and those who have never attended?

baseline

① We should generate a new dummy variable which takes the value of 1 if a student attended a choice school, and 0 otherwise.

→  $choice = \begin{cases} 1 & \text{if a student attended a choice school} \\ 0 & \text{have never attended} \end{cases}$

② We should run the following model :

$$mathscore = \beta_0 + \beta_1 choice + \beta_2 black + \beta_3 hispanic + \beta_4 female + u$$

difference in math score btw students who attended a choice school & tho who did not.

**Problem 4.** The following regression results are based on a dataset of working men from 1991, with *abil* representing cognitive ability and *educ* representing years of education.

. reg educ motheduc fatheduc abil abil2

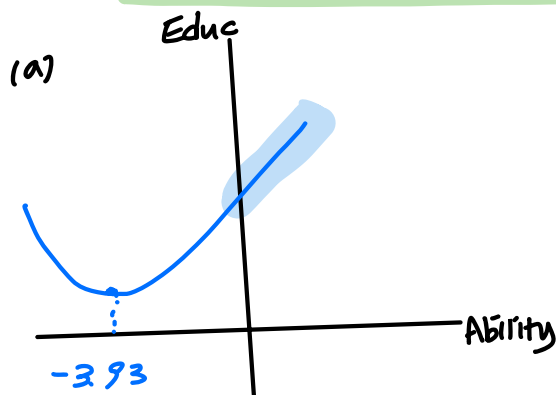
| Source   | SS         | df    | MS         | Number of obs | = | 1,230  |
|----------|------------|-------|------------|---------------|---|--------|
|          |            |       |            | F(4, 1225)    | = | 244.91 |
| Model    | 3027.03706 | 4     | 756.759264 | Prob > F      | = | 0.0000 |
| Residual | 3785.24262 | 1,225 | 3.08999397 | R-squared     | = | 0.4444 |
|          |            |       |            | Adj R-squared | = | 0.4425 |
| Total    | 6812.27967 | 1,229 | 5.54294522 | Root MSE      | = | 1.7578 |

| educ     | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |          |
|----------|-------------|-----------|-------|-------|----------------------|----------|
| motheduc | .1901261    | .0280957  | 6.77  | 0.000 | .1350051             | .2452472 |
| fatheduc | .1089387    | .0196014  | 5.56  | 0.000 | .0704827             | .1473946 |
| abil     | .4014624    | .0302875  | 13.26 | 0.000 | .3420413             | .4608835 |
| abil2    | .050599     | .0083039  | 6.09  | 0.000 | .0343076             | .0668905 |
| _cons    | 8.240226    | .2874099  | 28.67 | 0.000 | 7.676356             | 8.804097 |

$$\text{Educ} = \beta_0 + \beta_1 \text{abil} + \beta_2 \text{abil}^2 + \beta_3 \text{mothu} + \beta_4 \text{fedu} + u$$

- (a) Draw the relationship between cognitive ability (*abil*) and years of education (*educ*).
- (b) Find the turning point.
- (c) Conduct a significance test at the 5% level to evaluate whether the relationship between cognitive ability and years of education is linear.



(b) Turning pt: slope = 0

$$\frac{\partial \text{Educ}}{\partial \text{Abil}} = 0$$

$$\Rightarrow \beta_1 + 2\beta_2 \text{Abil} = 0$$

$$\text{Abil}^* = -\frac{\beta_1}{2\beta_2} = -\frac{0.401}{2(0.05)} = -3.93$$

not meaningful  
b/c ability is not negative  
(can't be)

(c) ①  $H_0: \beta_2 = 0$   
 $H_a: \beta_2 \neq 0$

② sig lv  $\alpha = 0.05$

③ test stat =  $\frac{\hat{\beta}_2 - \beta_{2,0}}{\text{se}(\hat{\beta}_2)} = \frac{\hat{\beta}_2}{\text{se}(\hat{\beta}_2)}$   
 $= \frac{0.050599}{0.0083039} = 6.09$

④ p-value =  $P(|t| > |\text{test-stat}|)$

$$= P(|t| > 6.09)$$

$$\approx 2 \times P(Z < -6.09) = 2 \times \phi = \phi$$

⑤ Conclusion

p-value of  $\phi$  is smaller than the sig. lv 0.05.

Therefore, we reject  $H_0$  that  $\beta_2 = 0$   
 $\therefore$  Educ & Ability have a nonlinear rel.

**Problem 5.** Using the same dataset, we run a different regression to examine the effect of number of siblings and college tuition. *sibs* is the number of siblings and *tuit17* is college tuition at age 17.

→ gen sib3 = sibs >= 3 ← dummy  $sib3 = \begin{cases} 1 & \text{if } \#sibs \geq 3 \\ 0 & \text{o/w} \end{cases}$

→ reg educ c.tuit17##sib3 = tuit17 sib3 tuit17 x sib3

| Source   | SS         | df    | MS         | Number of obs | = | 1,230  |
|----------|------------|-------|------------|---------------|---|--------|
| Model    | 305.529838 | 3     | 101.843279 | F(3, 1226)    | = | 19.19  |
| Residual | 6506.74984 | 1,226 | 5.30730003 | Prob > F      | = | 0.0000 |
|          |            |       |            | R-squared     | = | 0.0448 |
|          |            |       |            | Adj R-squared | = | 0.0425 |
| Total    | 6812.27967 | 1,229 | 5.54294522 | Root MSE      | = | 2.3038 |

|           | educ          | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |
|-----------|---------------|-------------|-----------|-------|-------|----------------------|
| $\beta_1$ | tuit17        | .0453057    | .0224781  | 2.02  | 0.044 | .0012059 .0894056    |
| $\beta_2$ | 1.sib3        | -.7144889   | .3028995  | -2.36 | 0.018 | -1.308748 -.1202302  |
| $\beta_3$ | sib3#c.tuit17 | -.0297482   | .0318676  | -0.93 | 0.351 | -.0922694 .0327729   |
| $\beta_0$ | _cons         | 13.16868    | .2116824  | 62.21 | 0.000 | 12.75338 13.58398    |

$$Educ = \beta_0 + \beta_1 tuit17 + \beta_2 sib3 + \beta_3 (tuit17 \times sib3) + u$$

- (a) Draw the relationship between years of education and college tuition at age 17 (*educ*) for those who have ~~more than three~~ <sup>or more</sup> siblings and those who have fewer ~~than three~~. Label the slope and intercept terms. (Ignore the statistical significance of the estimates for now.)

- (b) Can we conclude that the effect of college tuition at age 17 on years of education differs between those with more ~~than three~~ <sup>three or</sup> siblings and those with ~~three or~~ fewer? Explain.  $\Rightarrow \beta_3$

- (c) Redraw the relationship if necessary, based on your conclusion from (b).

No, the estimated coeff on the interaction term is not statistically sig. at the 5% lv.

**3+ siblings**

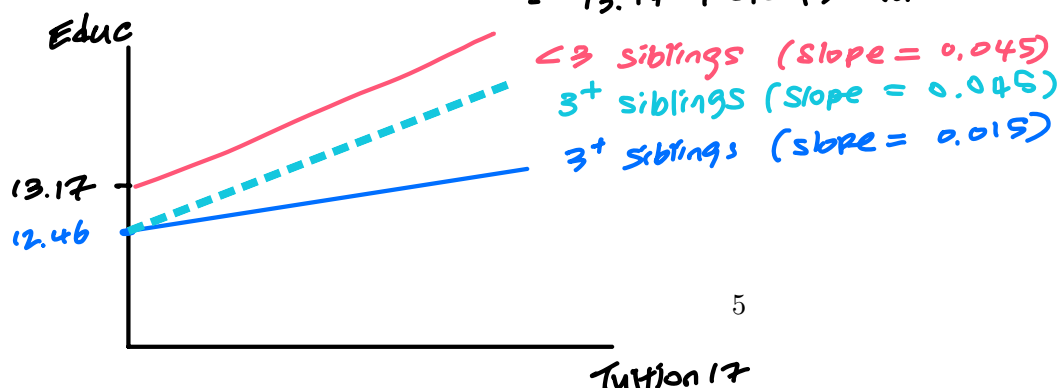
$$E[educ | tuit17, sib3=1] = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot tuit17$$

$$= 12.46 + 0.015 tuit17 \quad \checkmark$$

**<3 siblings**

$$E[educ | tuit17, sib3=0] = \beta_0 + \beta_1 tuit17$$

$$= 13.17 + 0.045 tuit17$$



$\Rightarrow$  3+ sib group  
and  
<3 sib group

have the same slope.

**STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.**

| <b>Z</b>    | <b>.00</b> | <b>.01</b> | <b>.02</b> | <b>.03</b> | <b>.04</b> | <b>.05</b> | <b>.06</b> | <b>.07</b> | <b>.08</b> | <b>.09</b> |
|-------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| <b>-3.9</b> | .00005     | .00005     | .00004     | .00004     | .00004     | .00004     | .00004     | .00004     | .00003     | .00003     |
| <b>-3.8</b> | .00007     | .00007     | .00007     | .00006     | .00006     | .00006     | .00006     | .00005     | .00005     | .00005     |
| <b>-3.7</b> | .00011     | .00010     | .00010     | .00010     | .00009     | .00009     | .00008     | .00008     | .00008     | .00008     |
| <b>-3.6</b> | .00016     | .00015     | .00015     | .00014     | .00014     | .00013     | .00013     | .00012     | .00012     | .00011     |
| <b>-3.5</b> | .00023     | .00022     | .00022     | .00021     | .00020     | .00019     | .00019     | .00018     | .00017     | .00017     |
| <b>-3.4</b> | .00034     | .00032     | .00031     | .00030     | .00029     | .00028     | .00027     | .00026     | .00025     | .00024     |
| <b>-3.3</b> | .00048     | .00047     | .00045     | .00043     | .00042     | .00040     | .00039     | .00038     | .00036     | .00035     |
| <b>-3.2</b> | .00069     | .00066     | .00064     | .00062     | .00060     | .00058     | .00056     | .00054     | .00052     | .00050     |
| <b>-3.1</b> | .00097     | .00094     | .00090     | .00087     | .00084     | .00082     | .00079     | .00076     | .00074     | .00071     |
| <b>-3.0</b> | .00135     | .00131     | .00126     | .00122     | .00118     | .00114     | .00111     | .00107     | .00104     | .00100     |
| <b>-2.9</b> | .00187     | .00181     | .00175     | .00169     | .00164     | .00159     | .00154     | .00149     | .00144     | .00139     |
| <b>-2.8</b> | .00256     | .00248     | .00240     | .00233     | .00226     | .00219     | .00212     | .00205     | .00199     | .00193     |
| <b>-2.7</b> | .00347     | .00336     | .00326     | .00317     | .00307     | .00298     | .00289     | .00280     | .00272     | .00264     |
| <b>-2.6</b> | .00466     | .00453     | .00440     | .00427     | .00415     | .00402     | .00391     | .00379     | .00368     | .00357     |
| <b>-2.5</b> | .00621     | .00604     | .00587     | .00570     | .00554     | .00539     | .00523     | .00508     | .00494     | .00480     |
| <b>-2.4</b> | .00820     | .00798     | .00776     | .00755     | .00734     | .00714     | .00695     | .00676     | .00657     | .00639     |
| <b>-2.3</b> | .01072     | .01044     | .01017     | .00990     | .00964     | .00939     | .00914     | .00889     | .00866     | .00842     |
| <b>-2.2</b> | .01390     | .01355     | .01321     | .01287     | .01255     | .01222     | .01191     | .01160     | .01130     | .01101     |
| <b>-2.1</b> | .01786     | .01743     | .01700     | .01659     | .01618     | .01578     | .01539     | .01500     | .01463     | .01426     |
| <b>-2.0</b> | .02275     | .02222     | .02169     | .02118     | .02068     | .02018     | .01970     | .01923     | .01876     | .01831     |
| <b>-1.9</b> | .02872     | .02807     | .02743     | .02680     | .02619     | .02559     | .02500     | .02442     | .02385     | .02330     |
| <b>-1.8</b> | .03593     | .03515     | .03438     | .03362     | .03288     | .03216     | .03144     | .03074     | .03005     | .02938     |
| <b>-1.7</b> | .04457     | .04363     | .04272     | .04182     | .04093     | .04006     | .03920     | .03836     | .03754     | .03673     |
| <b>-1.6</b> | .05480     | .05370     | .05262     | .05155     | .05050     | .04947     | .04846     | .04746     | .04648     | .04551     |
| <b>-1.5</b> | .06681     | .06552     | .06426     | .06301     | .06178     | .06057     | .05938     | .05821     | .05705     | .05592     |
| <b>-1.4</b> | .08076     | .07927     | .07780     | .07636     | .07493     | .07353     | .07215     | .07078     | .06944     | .06811     |
| <b>-1.3</b> | .09680     | .09510     | .09342     | .09176     | .09012     | .08851     | .08691     | .08534     | .08379     | .08226     |
| <b>-1.2</b> | .11507     | .11314     | .11123     | .10935     | .10749     | .10565     | .10383     | .10204     | .10027     | .09853     |
| <b>-1.1</b> | .13567     | .13350     | .13136     | .12924     | .12714     | .12507     | .12302     | .12100     | .11900     | .11702     |
| <b>-1.0</b> | .15866     | .15625     | .15386     | .15151     | .14917     | .14686     | .14457     | .14231     | .14007     | .13786     |
| <b>-0.9</b> | .18406     | .18141     | .17879     | .17619     | .17361     | .17106     | .16853     | .16602     | .16354     | .16109     |
| <b>-0.8</b> | .21186     | .20897     | .20611     | .20327     | .20045     | .19766     | .19489     | .19215     | .18943     | .18673     |
| <b>-0.7</b> | .24196     | .23885     | .23576     | .23270     | .22965     | .22663     | .22363     | .22065     | .21770     | .21476     |
| <b>-0.6</b> | .27425     | .27093     | .26763     | .26435     | .26109     | .25785     | .25463     | .25143     | .24825     | .24510     |
| <b>-0.5</b> | .30854     | .30503     | .30153     | .29806     | .29460     | .29116     | .28774     | .28434     | .28096     | .27760     |
| <b>-0.4</b> | .34458     | .34090     | .33724     | .33360     | .32997     | .32636     | .32276     | .31918     | .31561     | .31207     |
| <b>-0.3</b> | .38209     | .37828     | .37448     | .37070     | .36693     | .36317     | .35942     | .35569     | .35197     | .34827     |
| <b>-0.2</b> | .42074     | .41683     | .41294     | .40905     | .40517     | .40129     | .39743     | .39358     | .38974     | .38591     |
| <b>-0.1</b> | .46017     | .45620     | .45224     | .44828     | .44433     | .44038     | .43644     | .43251     | .42858     | .42465     |
| <b>-0.0</b> | .50000     | .49601     | .49202     | .48803     | .48405     | .48006     | .47608     | .47210     | .46812     | .46414     |

